INDEPENDENT UNIVERSITY (ULK)



UML CLASS DIAGRAM MODELLING ASSIGNMENT

PROJECT DESCRIPTION REPORT

Project Title: **Online Shopping System**
Instructor: Eric Maniraguha

Group Members

1. Student 1 : 202510585
2. Student 2 : 202512742

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1. Introduction

Online shopping has become an important part of modern business. It enables customers to view products, select items, place orders, and make payments without visiting a physical shop.

The Online Shopping System is a proposed business information system designed to represent the main activities involved in an online shopping process. The system includes customers, administrators, products, categories, shopping carts, orders, and payments.

This project focuses on modelling the system using a UML class diagram. The diagram represents the main classes, their attributes, operations, and relationships.

2. Problem Statement

Traditional shopping may require customers to visit physical shops to find products and complete purchases. Managing product information, customer details, shopping activities, orders, and payments can also become difficult when these activities are not properly organized.

An online shopping system provides a structured way to represent these activities. It helps organize information about products, customers, categories, carts, orders, and payments in one proposed system.

3. Project Objectives

3.1 General Objective

To design a UML class diagram for an Online Shopping System that represents its main classes, attributes, operations, and relationships.

3.2 Specific Objectives

- To identify the main classes required for the proposed system.
- To define appropriate attributes for each class.
- To identify the operations that each class can perform.
- To represent relationships between the classes.
- To specify multiplicities for the relationships.
- To document the system design using UML modelling principles.

4. Project Scope

The project focuses on the conceptual design of an Online Shopping System. It covers the following areas:

4.1 Customer Management

The Customer class represents people who use the system. It contains customer identification, name, telephone number, email address, and address. The class provides operations for registering and updating customer information.

4.2 Administrator Management

The Admin class represents an administrator responsible for managing aspects of the system, particularly product management.

4.3 Product Management

The Product class represents items available for purchase. It includes product identification, product name, and description. Its operations include updating product information and adding a product to a cart, as represented in the proposed diagram.

4.4 Category Management

The Category class represents groups used to organize products. It contains a category identifier and category name, with operations for adding and viewing categories.

4.5 Shopping Cart Management

The Cart class represents the customer's shopping cart. It contains customer and product identifiers and provides operations for updating and viewing cart information.

4.6 Order Management

The Order class represents orders placed by customers. It contains an order identifier and customer identifier, with operations for adding and viewing orders.

4.7 Payment Management

The Payment class represents payment-related information. The proposed diagram includes customer identification, cart type, customer telephone number, and card number, with operations for updating and viewing payment information.

5. Functional Requirements

The proposed system is expected to support the following functions:

1. Register customers.
2. Update customer information.
3. View available products.
4. Manage and update product information.
5. Add products to a shopping cart.
6. Update and view shopping cart information.
7. Add and view orders.
8. Add and view product categories.
9. Update and view payment-related information.
10. Allow administrators to manage products.

These requirements describe the intended functions of the proposed system; they do not mean that a working software application has already been implemented.

6. Non-Functional Requirements

The proposed design should support the following quality considerations:

- **Usability:** The system should be easy for customers and administrators to understand.
- **Reliability:** Information should be managed consistently.
- **Security:** Customer and payment information should be protected.
- **Maintainability:** Classes should be organized so that the design can be understood and improved.
- **Data integrity:** Relationships between customers, products, carts, orders, categories, and payments should be clearly defined.

7. Expected Benefits

The proposed Online Shopping System is expected to provide the following benefits:

- Organised representation of customers and products.
- Easier management of product categories.
- A structured representation of shopping carts and orders.
- Better organisation of payment-related information.
- Clearer understanding of how the main components of an online shopping business relate to one another.
- A foundation for future software implementation.

8. Project Limitations

This assignment focuses on UML class diagram modelling rather than the implementation of a complete online shopping application.

The current design does not explicitly define features such as inventory tracking, delivery management, refunds, order tracking, or customer authentication beyond the classes and operations shown in the diagram. These features could be considered in a future version if the project scope is expanded.

9. Conclusion

The Online Shopping System is a proposed model for organising important activities in an online shopping business. It identifies seven main classes: Admin, Customer, Product, Category, Cart, Order, and Payment.

The project applies UML class diagram modelling to describe the structure of the proposed system through classes, attributes, operations, and relationships. The resulting design provides a foundation for understanding the system and potentially implementing it in the future.